EcoRide Fleet Platform – Updated MVP Scope

Expanded Telematics & Vehicle Access Integration Scope (https://www.cancomsa.com/canfleet-sa/)

The MVP scope has now been expanded to include **basic telematics platform integration and controlled vehicle access workflows**. This expands the original MVP from a purely operational onboarding and scheduling platform into an integrated operational fleet-control platform.

Revised MVP Positioning

Updated Executive Summary

The EcoRide MVP is designed to digitise the core operational loop for a shift-based e-hailing fleet while introducing foundational telematics integration capabilities.

In addition to onboarding, shift scheduling, notifications, and earnings visibility, the MVP will now include:

- Driver registration synchronisation with the telematics platform
- Driver-to-vehicle assignment integration
- Controlled vehicle access workflows
- Vehicle unlock authorisation
- Camera activation workflows
- Biometric ignition validation orchestration

This establishes the foundation for future:

- Full fleet control
- AI-driven fleet optimisation

- Real-time telematics analytics
- Vehicle security automation
- Driver behaviour scoring

Updated MVP Scope

Area	Included in MVP	Excluded from MVP
Driver onboarding	Registration, profile capture, required docs, agreement signing, application submission	Advanced biometric onboarding
Shift operations	Shift creation, requests, confirmations, reminders	AI allocation logic
Financial visibility	Earnings publishing and visibility	Automated payouts
Telematics integration	Driver registration sync, driver-vehicle assignment sync, unlock requests, vehicle access validation	Full real-time telemetry analytics
Vehicle access control	Controlled unlock workflows, ignition orchestration, biometric validation trigger orchestration	Advanced remote immobilisation automation
Vehicle security	Driver assignment validation and biometric ignition check integration	Automated hijack response
Fleet control	Basic telematics integration	Live GPS fleet tracking dashboards
AI functionality	None in MVP	Heatmaps, AI routing, predictive analytics

New MVP Capability: Telematics Integration

Objective

Provide a lightweight but production-capable integration between the EcoRide platform and the telematics provider to:

- Register approved drivers
- Allocate drivers to vehicles and shifts
- Allow authorised drivers to unlock vehicles
- Enforce driver validation before ignition
- Establish the operational foundation for future advanced fleet management

Updated Product Architecture

Existing Layers

Mobile App

Driver-facing application.

Web App

Operations/admin dashboard.

Shared Backend

Workflow orchestration, validation, integrations, and business logic.

NEW: External Telematics Platform Integration Layer

A new integration layer will connect EcoRide to the telematics provider via REST APIs.

Responsibilities

- Vehicle registration synchonisation
- Driver registration synchronisation
- Vehicle assignment synchronisation
- Shift access control
- Unlock request orchestration
- Vehicle access audit trail
- Biometric validation orchestration
- Event/status callbacks

Updated Shared Backend Scope

The backend will now additionally:

- Integrate with telematics APIs
- Maintain telematics sync status
- Validate shift-based vehicle access
- Orchestrate unlock requests
- Handle ignition validation workflows
- Store telematics integration audit logs

New MVP Operational Workflow

1. Driver Onboarding Workflow

Existing Process

- Driver registers
- Uploads documents
- Signs agreement
- Submits application
- Admin approves

NEW Telematics Integration Step

Once approved:

- EcoRide backend sends approved driver record to telematics platform
- Telematics platform creates/registers driver identity
- Driver telematics ID returned and stored

New Backend Fields

Add to `DriverProfiles`:

- telematics_driver_id
- telematics_sync_status
- telematics_registered_at

2. Shift Allocation & Vehicle Assignment Workflow

New Workflow

When operations allocates:

- Driver
- Vehicle
- Shift window

The EcoRide backend:

1. Creates allocation internally
2. Sends assignment to telematics platform
3. Registers:
    a. Driver ID
    b. Vehicle ID
    c. Shift start/end time
4. Telematics platform validates authorised usage window

NEW Collection: VehicleAssignments

Field	Purpose
driver_profile_id	Assigned driver
vehicle_id	Assigned vehicle
shift_id	Related shift
assignment_start	Allowed start
assignment_end	Allowed end
telematics_assignment_id	External assignment reference
status	active/completed/cancelled
created_at	Audit timestamp

3. Vehicle Unlock Workflow

New MVP Capability

The driver app will now support:

- Unlock vehicle action

Unlock Flow

1. Driver logs into EcoRide app
2. Driver has:
    a. Approved profile
    b. Active shift
    c. Assigned vehicle
3. Driver presses “Unlock Vehicle”
4. EcoRide backend sends unlock request to telematics platform
5. Telematics system validates:
    a. Driver exists
    b. Driver assigned to vehicle
    c. Shift currently active
6. If valid:
    a. Vehicle doors unlock

4. Ignition & Biometric Workflow

MVP Integration Behaviour

After unlock:

- Vehicle ignition accessory position activates
- Camera system powers on
- Telematics/camera platform performs biometric validation

Validation Flow

1. Driver enters vehicle
2. Ignition accessory activated
3. Camera captures driver image
4. Telematics platform validates:
    a. Facial match
    b. Driver assignment
    c. Valid shift window
5. If successful:

    a. Ignition enabled
6. If failed:
    a. Ignition blocked
    b. Failure event logged

Important MVP Boundary

The MVP will:

- Integrate with biometric validation workflows
- Trigger validation processes
- Receive pass/fail results

The MVP will NOT:

- Build biometric AI internally
- Store biometric templates
- Implement custom facial recognition models

These remain responsibilities of the telematics/camera platform.

Updated Mobile App Scope

New Driver Dashboard Features

Vehicle Access Card

Displays:

- Assigned vehicle
- Shift time
- Vehicle status
- Unlock button

New Mobile App Features

Feature	Description
Unlock Vehicle	Sends unlock request
Vehicle Access Status	Shows assigned vehicle state
Shift Vehicle Visibility	Displays active vehicle allocation
Unlock Failure Feedback	Displays reason for denial
Telematics Status	Optional sync/connection status

Updated Shared Backend Collections

New Collection: Vehicles

Field	Purpose
vehicle_id	Internal vehicle ID
registration_number	Vehicle registration
telematics_vehicle_id	External telematics reference
status	active/inactive/service
telematics_enabled	Boolean flag

New Collection: VehicleAccessEvents

Field	Purpose
driver_profile_id	Driver attempting access
vehicle_id	Vehicle accessed
event_type	unlock_request / ignition_attempt / biometric_pass / biometric_fail
event_status	success/failure
event_timestamp	Audit timestamp

source_reference Telematics event ID

Updated Workflow Automations

New Workflow: Driver Telematics Registration

Trigger

Driver approved.

Behaviour

- Register driver in telematics platform
- Store telematics reference ID
- Update sync status
- Retry on failure

New Workflow: Vehicle Assignment Sync

Trigger

Driver allocated to vehicle/shift.

Behaviour

- Send assignment to telematics platform
- Activate permitted access window
- Store assignment confirmation

New Workflow: Vehicle Unlock Validation

Trigger

Driver presses unlock.

Behaviour

- Validate assignment
- Validate shift timing
- Validate active status
- Send unlock request
- Record result

New Workflow: Ignition Validation

Trigger

Ignition accessory activated.

Behaviour

- Trigger biometric validation
- Await telematics response
- Permit or deny ignition
- Log result

Updated Validation Rules

New Validation Rules

Vehicle Unlock Requires

- Approved driver

- Signed agreement
- Active shift
- Active assignment
- Valid telematics registration
- Shift time within allowed window

Ignition Requires

- Successful unlock
- Biometric pass
- Matching assigned driver

Updated MVP Exclusions

The following remain OUT OF SCOPE for MVP:

- AI route optimisation
- Heatmaps and predictive demand
- Automated hijack response
- Real-time live tracking dashboards
- Driver scoring
- Advanced telematics analytics
- Automated vehicle immobilisation rules
- Full telemetry streaming/data lake implementation
- Predictive maintenance
- Vehicle damage AI assessment
- Fuel optimisation analytics

Updated Completion Criteria

The MVP should now additionally support the following end-to-end operational loop:

1. Driver registers in EcoRide app.
2. Driver completes onboarding and approval.

3. Driver is synchronised to telematics platform.
4. Admin allocates driver to:
    a. Shift
    b. Vehicle
5. Assignment synchronises to telematics platform.
6. Driver opens EcoRide app and requests vehicle unlock.
7. Telematics platform validates assignment and unlocks vehicle.
8. Ignition accessory activates camera system.
9. Biometric validation occurs.
10. Vehicle ignition enabled only if validation passes.
11. All access events logged successfully.

Architectural Outcome

The updated MVP now establishes:

- A unified operational platform
- Foundational telematics integration
- Secure vehicle access control
- Shift-based fleet governance
- Driver identity enforcement
- Scalable API-first fleet architecture

while still remaining lightweight enough for rapid MVP delivery and pilot deployment.